WORKED SOLUTION

STEP II 2020, Question 7

Pure •

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Read the question before you go on. It is in the STEP II paper in OCR's download of the 2020 STEP papers (a zip file):

www.ocr.org.uk/Images/696589-2020-past-papers.zip

The key idea

The map $z \mapsto w = \dfrac{2}{z-2}$ is a shift followed by an inversion, and inversion sends lines that miss the origin to circles through the origin. To *prove* that w lies on a particular circle, the cleanest route is to guess the centre c and then show that $|w-c|$ simplifies to a constant. Writing $w-c$ as a single fraction, the numerator and denominator turn out to be conjugates (or nearly so), which is what makes the modulus constant.

Solution

Throughout we use $|\alpha/\beta| = |\alpha|/|\beta|$ and $|x+yi| = |x-yi| = |y+xi| = \sqrt{x^2+y^2}$ for real x, y.

Part (i)

With $z = 3+ti$ we have $z-2 = 1+ti$, so

$$w - 1 = \frac{2}{1+ti} - 1 = \frac{2-(1+ti)}{1+ti} = \frac{1-ti}{1+ti}.$$

The numerator and denominator are complex conjugates, so they have the same modulus, $\sqrt{1+t^2}$, which is never zero. Hence

$$|w-1| = \frac{\sqrt{1+t^2}}{\sqrt{1+t^2}} = 1,$$

which does not depend on t. Every z on the line $\mathrm{Re}(z) = 3$ has the form $3+ti$ for some real t, so for every such z the point w is at distance 1 from 1: it lies on the circle with centre 1 and radius 1.

The line V. Now $z = p+ti$, and it helps to write $a = p-2$, which is a non-zero real number because $p \neq 2$. Then $z-2 = a+ti$, which is never zero. When $p = 3$ we had

centre $1 = 1/a$, so we try the centre $1/a$ in general:

$$w - \frac{1}{a} = \frac{2}{a+ti} - \frac{1}{a} = \frac{2a-(a+ti)}{a(a+ti)} = \frac{a-ti}{a(a+ti)}.$$

Again $|a - ti| = |a + ti|$, so

$$\left|w - \frac{1}{p-2}\right| = \frac{1}{|a|} = \frac{1}{|p-2|}.$$

So w lies on the circle with

$$\text{centre } \frac{1}{p-2}, \qquad \text{radius } \frac{1}{|p-2|}.$$

The modulus sign in the radius matters: if $p < 2$ the centre is on the negative real axis, but a radius must be positive.

Ask yourself How would we have found the centre $1/(p-2)$ if we hadn't been told the $p = 3$ case?

The point $z = p$ (where $t = 0$) gives $w = 2/a$, and as $t \to \pm\infty$ we get $w \to 0$. These two points are at opposite ends of a diameter (the circle is symmetric about the real axis, and they are its two real points), so the centre is their midpoint, $1/a$.

For the imaginary part, multiply top and bottom by the conjugate:

$$w = \frac{2(a-ti)}{a^2+t^2}, \qquad \text{so} \qquad \mathrm{Im}(w) = \frac{-2t}{a^2+t^2}.$$

The denominator is positive, so $\mathrm{Im}(w) > 0$ exactly when $t < 0$. Hence the points of V with $\mathrm{Im}(w) > 0$ are

$$z = p + ti \text{ with } t < 0, \quad \text{that is, the points of } V \text{ below the real axis } (\mathrm{Im}(z) < 0).$$

Part (ii)

Now $z = s + qi$ with s real, and we write $u = s - 2$, so $z - 2 = u + qi$, which is non-zero since $q \neq 0$. Here the two 'ends' of the circle are $w = 0$ (as $u \to \pm\infty$) and $w = 2/(qi) = -2i/q$ (at $u = 0$), suggesting the centre $-i/q$. Check:

$$w + \frac{i}{q} = \frac{2}{u+qi} + \frac{i}{q} = \frac{2q + i(u+qi)}{q(u+qi)} = \frac{q+ui}{q(u+qi)}.$$

Now $|q + ui| = \sqrt{q^2+u^2} = |u + qi|$, so

$$\left|w + \frac{i}{q}\right| = \frac{1}{|q|}.$$

So w lies on the circle with

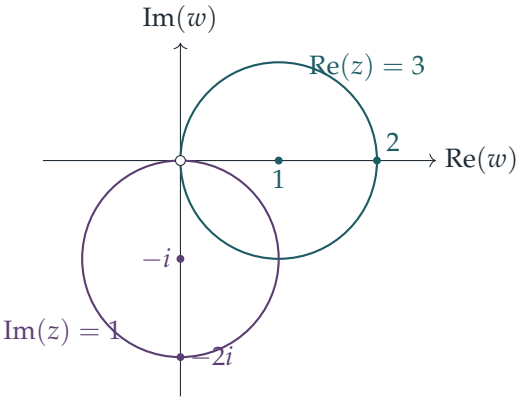
$$\text{centre } -\frac{i}{q}, \qquad \text{radius } \frac{1}{|q|}.$$

For the real part,

$$w = \frac{2(u - qi)}{u^2 + q^2}, \qquad \text{so} \qquad \mathrm{Re}(w) = \frac{2u}{u^2 + q^2},$$

with positive denominator. So $\mathrm{Re}(w) > 0$ exactly when $u > 0$, that is, for

$$z = s + qi \text{ with } s > 2, \quad \text{the points of } H \text{ to the right of } \mathrm{Re}(z) = 2.$$



The figure shows the image of $\mathrm{Re}(z) = 3$ (from part (i)) and of $\mathrm{Im}(z) = 1$ (part (ii) with $q = 1$). Both circles pass through the origin, but $w = 0$ itself is never reached, since $2/(z - 2) \neq 0$; it corresponds to z 'at infinity' along the line.

Checks

- Take $p = 3$ in the formula for V: centre $1/(3 - 2) = 1$ and radius 1, matching the first result.
- Spot values: $z = 3 - i$ gives $w = 2/(1 - i) = 1 + i$, which has $|w - 1| = 1$ and $\mathrm{Im}(w) = 1 > 0$, consistent with $t = -1 < 0$. For H with $q = 1$, $z = 3 + i$ gives $w = 1 - i$, which is at distance 1 from $-i$ and has positive real part, as $3 > 2$.
- The radius $1/|p - 2|$ grows as $p \to 2$: the line V then passes close to the point $z = 2$ that is sent to infinity, so its image should be huge. A computer check over thousands of random p, q and points on each line confirmed both circles and both sign conditions.

Common slips

- Giving the radius as $1/(p - 2)$ or $1/q$, which is negative when $p < 2$ or $q < 0$.
- Getting the sign of $\mathrm{Im}(w)$ the wrong way round: dividing by $z - 2$ reflects in the real axis, so $\mathrm{Im}(w)$ has the opposite sign to $\mathrm{Im}(z)$.

Beyond STEP

The map $z \mapsto 2/(z-2)$ is a Möbius transformation, $z \mapsto (az+b)/(cz+d)$. These maps always send 'generalised circles' (circles and lines) to generalised circles, and they are the basic geometric tool of a first course in complex analysis, from conformal mapping to hyperbolic geometry. Needham's *Visual Complex Analysis* has a beautiful chapter on them, with pictures that make results like this one obvious.

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